

Two-Column Research Note

Introduction. Retrieval quality depends heavily on chunk boundaries. Sentences that are split across chunks lose context, while chunks that are too large dilute relevance scoring. This note evaluates three chunking strategies on a held-out corpus of technical documents. Prior work on retrieval-augmented generation has largely treated chunking as a fixed preprocessing step rather than a tunable parameter of the overall system. We define a chunk boundary as correct if it does not split a sentence that a human annotator judged to be a single semantic unit. The held-out corpus spans technical manuals, financial filings, and scientific papers to avoid overfitting the comparison to a single document genre. Fixed-width chunking was implemented with a sliding window and no sentence awareness, serving as the baseline for every comparison in this note. Paragraph-based chunking preserved more local context but produced highly variable chunk sizes, complicating downstream token budgeting. We measured both retrieval recall and the rate of citations that pointed to the wrong sentence within an otherwise correctly retrieved chunk. Annotators were shown candidate chunk boundaries without knowing which strategy produced them, to avoid biasing the manual review. The evaluation set was frozen before any chunking strategy was implemented, to prevent the boundary definitions from being tuned to the test data. Chunk overlap was varied independently of chunk size to isolate its effect on boundary-loss recovery. All three strategies were run against the same embedding model and retrieval index to keep the comparison isolated to chunking behavior alone.

Results. The sentence-aware strategy outperformed fixed-width chunking by 8 points of recall at $k=10$. Overlap of one sentence between adjacent chunks recovered most of the boundary-loss cases without materially increasing token cost. A secondary analysis restricted to tables and figures showed a smaller but still significant advantage for the sentence-aware strategy. Latency overhead from sentence segmentation was under five milliseconds per page, well within the ingestion pipeline's per-page budget. The fixed-width baseline's failures clustered heavily around numbered lists and multi-clause sentences with embedded citations. Manual review of a sample of one hundred citations found no cases where the sentence-aware strategy introduced a new failure mode. We did not observe a meaningful difference in indexing cost between the strategies, since token count dominates cost far more than segmentation choice. Future work should evaluate whether a learned boundary model outperforms the rules-based sentence splitter used here. The one-sentence overlap setting was chosen after a small sweep over zero, one, and two sentences of overlap on a validation split. Two-sentence overlap recovered marginally more boundary-loss cases but increased average chunk size beyond the target token range. We recommend the sentence-aware strategy with one-sentence overlap as the default for prose-heavy technical corpora going forward. This note does not address chunking for pages dominated by tables, which we treat as a separate problem in a companion study. A follow-up experiment measured wall-clock indexing time across all three strategies on a shared hardware configuration. The sentence-aware strategy added roughly three percent to total indexing time, a cost we consider acceptable given the recall improvement. We also examined whether chunk overlap should vary by document type rather than using a single fixed value across the corpus. Preliminary results suggest technical manuals benefit from slightly larger overlap than financial filings, though the sample size here is small.